

NetflixBusinessCaseStudy

May 6, 2026

1 1. Problem Statement

- Netflix is a global media streaming company, with over 10,000 movies, TV shows and regional content available in multiple languages
- Netflix has over 222 Million subscribers globally.
- Netflix want to explore possible areas of opportunities to increase its reach and penetration in different countries and hence increase revenue.
- In addition to this, we want to identify unsuccessful ventures to prevent leakages and improve profits.
- The ultimate aim is to present analysis of netflix data in order to support decision for business growth in different countries.
- We want to suggest which type of shows to produce to grow the business
- The exploration should have a goal.
- As you explore the data, keep in mind that you want to answer which type of shows to produce and how to grow the business.
- Ensure each recommendation is backed by data. The company is looking for data-driven insights, not personal opinions or anecdotes.
- Assume that you are presenting your findings to business executives who have only a basic understanding of data science. Avoid unnecessary technical jargon.
- Start by exploring a few questions:
 - What type of content is available in different countries?
 - How has the number of movies released per year changed over the last 20-30 years?
 - Comparison of tv shows vs. movies.
 - What is the best time to launch a TV show?
 - Analysis of actors/directors of different types of shows/movies.
 - Does Netflix has more focus on TV Shows than movies in recent years
 - Understanding what content is available in different countries

2 1. Exploring Data

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[2]: netflix = pd.read_csv('netflix.csv')
```

```
[3]: netflix.head(3)
```

```
[3]:  show_id    type          title    director \
0      s1      Movie  Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show      Blood & Water          NaN
2      s3  TV Show      Ganglands  Julien Leclercq

                                cast          country \
0                                NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...  South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...    NaN

    date_added  release_year  rating  duration \
0  September 25, 2021        2020  PG-13    90 min
1  September 24, 2021        2021  TV-MA  2 Seasons
2  September 24, 2021        2021  TV-MA    1 Season

                                listed_in \
0                                Documentaries
1  International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act...

                                description
0  As her father nears the end of his life, filmm...
1  After crossing paths at a party, a Cape Town t...
2  To protect his family from a powerful drug lor...
```

```
[4]: netflix.set_index('show_id', inplace=True)
```

2.0.1 1.1 Data Preparation and Cleaning

Shape of Data, data types of attributes, Missing Values etc

```
[5]: netflix.shape
```

```
[5]: (8807, 11)
```

```
[6]: netflix.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 8807 entries, s1 to s8807
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   type            8807 non-null  object
1   title           8807 non-null  object
2   director        6173 non-null  object
3   cast            7982 non-null  object
4   country         7976 non-null  object
```

```

5   date_added      8797 non-null   object
6   release_year    8807 non-null   int64
7   rating          8803 non-null   object
8   duration        8804 non-null   object
9   listed_in       8807 non-null   object
10  description     8807 non-null   object
dtypes: int64(1), object(10)
memory usage: 825.7+ KB

```

Here we can observe that all columns except `release_year` is in string format. So we need to change the data types of some of the columns

- Checking the number of missing Values

```
[7]: netflix.isna().sum()
```

```

[7]: type          0
    title          0
    director      2634
    cast          825
    country       831
    date_added     10
    release_year   0
    rating         4
    duration       3
    listed_in      0
    description    0
    dtype: int64

```

```

[8]: # Checking the rows having missing values for specific columns ['date_added',
    ↪ 'rating', 'duration']
    netflix[netflix[['date_added', 'rating', 'duration']].isna().any(axis=1)]

```

```

[8]:      type      title \
show_id
s5542   Movie      Louis C.K. 2017
s5795   Movie      Louis C.K.: Hilarious
s5814   Movie      Louis C.K.: Live at the Comedy Store
s5990   Movie  13TH: A Conversation with Oprah Winfrey & Ava ...
s6067   TV Show      A Young Doctor's Notebook and Other Stories
s6175   TV Show      Anthony Bourdain: Parts Unknown
s6796   TV Show      Frasier
s6807   TV Show      Friends
s6828   TV Show      Gargantia on the Verdurous Planet
s6902   TV Show      Gunslinger Girl
s7197   TV Show      Kikoriki
s7255   TV Show      La Familia P. Luche
s7313   TV Show      Little Lunch
s7407   TV Show      Maron

```

s7538	Movie	My Honor Was Loyalty
s7848	TV Show	Red vs. Blue
s8183	TV Show	The Adventures of Figaro Pho

	director	cast \
show_id		
s5542	Louis C.K.	Louis C.K.
s5795	Louis C.K.	Louis C.K.
s5814	Louis C.K.	Louis C.K.
s5990	NaN	Oprah Winfrey, Ava DuVernay
s6067	NaN	Daniel Radcliffe, Jon Hamm, Adam Godley, Chris...
s6175	NaN	Anthony Bourdain
s6796	NaN	Kelsey Grammer, Jane Leeves, David Hyde Pierce...
s6807	NaN	Jennifer Aniston, Courteney Cox, Lisa Kudrow, ...
s6828	NaN	Kaito Ishikawa, Hisako Kanemoto, Ai Kayano, Ka...
s6902	NaN	Yuuka Nanri, Kanako Mitsuhashi, Eri Sendai, Am...
s7197	NaN	Igor Dmitriev
s7255	NaN	Eugenio Derbez, Consuelo Duval, Luis Manuel Áv...
s7313	NaN	Flynn Curry, Olivia Deeble, Madison Lu, Oisín ...
s7407	NaN	Marc Maron, Judd Hirsch, Josh Brener, Nora Zeh...
s7538	Alessandro Pepe	Leone Frisa, Paolo Vaccarino, Francesco Miglio...
s7848	NaN	Burnie Burns, Jason Saldaña, Gustavo Sorola, G...
s8183	NaN	Luke Jurevicius, Craig Behenna, Charlotte Haml...

	country	date_added	release_year	rating	duration \
show_id					
s5542	United States	April 4, 2017	2017	74 min	NaN
s5795	United States	September 16, 2016	2010	84 min	NaN
s5814	United States	August 15, 2016	2015	66 min	NaN
s5990	NaN	January 26, 2017	2017	NaN	37 min
s6067	United Kingdom	NaN	2013	TV-MA	2 Seasons
s6175	United States	NaN	2018	TV-PG	5 Seasons
s6796	United States	NaN	2003	TV-PG	11 Seasons
s6807	United States	NaN	2003	TV-14	10 Seasons
s6828	Japan	December 1, 2016	2013	NaN	1 Season
s6902	Japan	NaN	2008	TV-14	2 Seasons
s7197	NaN	NaN	2010	TV-Y	2 Seasons
s7255	United States	NaN	2012	TV-14	3 Seasons
s7313	Australia	February 1, 2018	2015	NaN	1 Season
s7407	United States	NaN	2016	TV-MA	4 Seasons
s7538	Italy	March 1, 2017	2015	NaN	115 min
s7848	United States	NaN	2015	NR	13 Seasons
s8183	Australia	NaN	2015	TV-Y7	2 Seasons

	listed_in \
show_id	
s5542	Movies

s5795	Movies
s5814	Movies
s5990	Movies
s6067	British TV Shows, TV Comedies, TV Dramas
s6175	Docuseries
s6796	Classic & Cult TV, TV Comedies
s6807	Classic & Cult TV, TV Comedies
s6828	Anime Series, International TV Shows
s6902	Anime Series, Crime TV Shows
s7197	Kids' TV
s7255	International TV Shows, Spanish-Language TV Sh...
s7313	Kids' TV, TV Comedies
s7407	TV Comedies
s7538	Dramas
s7848	TV Action & Adventure, TV Comedies, TV Sci-Fi ...
s8183	Kids' TV, TV Comedies

	description
show_id	
s5542	Louis C.K. muses on religion, eternal love, gi...
s5795	Emmy-winning comedy writer Louis C.K. brings h...
s5814	The comic puts his trademark hilarious/thought...
s5990	Oprah Winfrey sits down with director Ava DuVe...
s6067	Set during the Russian Revolution, this comic ...
s6175	This CNN original series has chef Anthony Bour...
s6796	Frasier Crane is a snooty but lovable Seattle ...
s6807	This hit sitcom follows the merry misadventure...
s6828	After falling through a wormhole, a space-dwel...
s6902	On the surface, the Social Welfare Agency appe...
s7197	A wacky rabbit and his gang of animal pals hav...
s7255	This irreverent sitcom featues Ludovico, Feder...
s7313	Adopting a child's perspective, this show take...
s7407	Marc Maron stars as Marc Maron, who interviews...
s7538	Amid the chaos and horror of World War II, a c...
s7848	This parody of first-person shooter games, mil...
s8183	Imagine your worst fears, then multiply them: ...

We can observe that the rating column for indexes s5542, s5795, s5814 contains the value of duration.
- so we can fill the empty duration values for these rows and search online for their rating value.

```
[9]: # entering the rating values to duration values
indexes = netflix[netflix['duration'].isna()].index
```

```
[10]: indexes
```

```
[10]: Index(['s5542', 's5795', 's5814'], dtype='object', name='show_id')
```

```
[11]: netflix.loc[indexes, 'duration'] = netflix.loc[indexes, 'rating']
```

```
[12]: netflix.loc[indexes, 'rating'] = np.nan
```

```
[13]: netflix.loc[indexes]
```

```
[13]:
```

	type		title	director	cast	\
show_id						
s5542	Movie		Louis C.K. 2017	Louis C.K.	Louis C.K.	
s5795	Movie		Louis C.K.: Hilarious	Louis C.K.	Louis C.K.	
s5814	Movie		Louis C.K.: Live at the Comedy Store	Louis C.K.	Louis C.K.	

	country	date_added	release_year	rating	duration	\
show_id						
s5542	United States	April 4, 2017	2017	NaN	74 min	
s5795	United States	September 16, 2016	2010	NaN	84 min	
s5814	United States	August 15, 2016	2015	NaN	66 min	

	listed_in		description
show_id			
s5542	Movies		Louis C.K. muses on religion, eternal love, gi...
s5795	Movies		Emmy-winning comedy writer Louis C.K. brings h...
s5814	Movies		The comic puts his trademark hilarious/thought...

```
[14]: netflix.isna().sum()
```

```
[14]: type          0
      title         0
      director    2634
      cast        825
      country     831
      date_added   10
      release_year  0
      rating       7
      duration     0
      listed_in    0
      description  0
      dtype: int64
```

```
[15]: # Now the rating columns have 7 missing values, we can just enter the correct_
      ↪value based on some online search for that title
```

```
[16]: indexes = netflix[netflix['rating'].isna()].index
```

```
[17]: indexes
```

```
[17]: Index(['s5542', 's5795', 's5814', 's5990', 's6828', 's7313', 's7538'],
      dtype='object', name='show_id')
```

```
[18]: netflix.loc[indexes]
```

```

[18]:
show_id      type      title \
s5542      Movie      Louis C.K. 2017
s5795      Movie      Louis C.K.: Hilarious
s5814      Movie      Louis C.K.: Live at the Comedy Store
s5990      Movie      13TH: A Conversation with Oprah Winfrey & Ava ...
s6828      TV Show      Gargantia on the Verdurous Planet
s7313      TV Show      Little Lunch
s7538      Movie      My Honor Was Loyalty

show_id      director      cast \
s5542      Louis C.K.      Louis C.K.
s5795      Louis C.K.      Louis C.K.
s5814      Louis C.K.      Louis C.K.
s5990      NaN      Oprah Winfrey, Ava DuVernay
s6828      NaN      Kaito Ishikawa, Hisako Kanemoto, Ai Kayano, Ka...
s7313      NaN      Flynn Curry, Olivia Deeble, Madison Lu, Oisín ...
s7538      Alessandro Pepe      Leone Frisa, Paolo Vaccarino, Francesco Miglio...

show_id      country      date_added      release_year      rating      duration \
s5542      United States      April 4, 2017      2017      NaN      74 min
s5795      United States      September 16, 2016      2010      NaN      84 min
s5814      United States      August 15, 2016      2015      NaN      66 min
s5990      NaN      January 26, 2017      2017      NaN      37 min
s6828      Japan      December 1, 2016      2013      NaN      1 Season
s7313      Australia      February 1, 2018      2015      NaN      1 Season
s7538      Italy      March 1, 2017      2015      NaN      115 min

show_id      listed_in \
s5542      Movies
s5795      Movies
s5814      Movies
s5990      Movies
s6828      Anime Series, International TV Shows
s7313      Kids' TV, TV Comedies
s7538      Dramas

show_id      description
s5542      Louis C.K. muses on religion, eternal love, gi...
s5795      Emmy-winning comedy writer Louis C.K. brings h...
s5814      The comic puts his trademark hilarious/thought...
s5990      Oprah Winfrey sits down with director Ava DuVe...
s6828      After falling through a wormhole, a space-dwel...

```

```
s7313    Adopting a child's perspective, this show take...
s7538    Amid the chaos and horror of World War II, a c...
```

```
[19]: netflix[netflix['type']=='Movie']['rating'].unique()
```

```
[19]: array(['PG-13', 'PG', 'TV-MA', 'TV-PG', 'TV-14', 'TV-Y', 'R', 'TV-G',
          'TV-Y7', 'G', 'NC-17', nan, 'NR', 'TV-Y7-FV', 'UR'], dtype=object)
```

```
[20]: replacement = pd.Series(['TV-MA', 'TV-MA', 'TV-MA', 'TV-PG', 'TV-14', 'TV-MA',
                               ↪ 'PG-13'], index=indexes)
```

```
[21]: replacement
```

```
[21]: show_id
s5542    TV-MA
s5795    TV-MA
s5814    TV-MA
s5990    TV-PG
s6828    TV-14
s7313    TV-MA
s7538    PG-13
dtype: object
```

```
[22]: netflix.loc[indexes, 'rating'] = replacement
```

```
[23]: netflix.loc[indexes]
```

```
[23]:
```

	show_id	type	title \
	s5542	Movie	Louis C.K. 2017
	s5795	Movie	Louis C.K.: Hilarious
	s5814	Movie	Louis C.K.: Live at the Comedy Store
	s5990	Movie	13TH: A Conversation with Oprah Winfrey & Ava ...
	s6828	TV Show	Gargantia on the Verdurous Planet
	s7313	TV Show	Little Lunch
	s7538	Movie	My Honor Was Loyalty

	show_id	director	cast \
	s5542	Louis C.K.	Louis C.K.
	s5795	Louis C.K.	Louis C.K.
	s5814	Louis C.K.	Louis C.K.
	s5990	NaN	Oprah Winfrey, Ava DuVernay
	s6828	NaN	Kaito Ishikawa, Hisako Kanemoto, Ai Kayano, Ka...
	s7313	NaN	Flynn Curry, Olivia Deeble, Madison Lu, Oisín ...
	s7538	Alessandro Pepe	Leone Frisa, Paolo Vaccarino, Francesco Miglio...

	country	date_added	release_year	rating	duration \
--	---------	------------	--------------	--------	------------

show_id						
s5542	United States	April 4, 2017	2017	TV-MA	74 min	
s5795	United States	September 16, 2016	2010	TV-MA	84 min	
s5814	United States	August 15, 2016	2015	TV-MA	66 min	
s5990	NaN	January 26, 2017	2017	TV-PG	37 min	
s6828	Japan	December 1, 2016	2013	TV-14	1 Season	
s7313	Australia	February 1, 2018	2015	TV-MA	1 Season	
s7538	Italy	March 1, 2017	2015	PG-13	115 min	

	listed_in \
show_id	
s5542	Movies
s5795	Movies
s5814	Movies
s5990	Movies
s6828	Anime Series, International TV Shows
s7313	Kids' TV, TV Comedies
s7538	Dramas

	description
show_id	
s5542	Louis C.K. muses on religion, eternal love, gi...
s5795	Emmy-winning comedy writer Louis C.K. brings h...
s5814	The comic puts his trademark hilarious/thought...
s5990	Oprah Winfrey sits down with director Ava DuVe...
s6828	After falling through a wormhole, a space-dwel...
s7313	Adopting a child's perspective, this show take...
s7538	Amid the chaos and horror of World War II, a c...

```
[24]: netflix.isna().sum()
```

```
[24]: type          0
      title          0
      director    2634
      cast        825
      country     831
      date_added   10
      release_year    0
      rating        0
      duration      0
      listed_in     0
      description    0
      dtype: int64
```

Now We have cleared the missing values from Duration and ratings columns

```
[26]: # Converting date_added column to datetime
      netflix['date_added'] = pd.to_datetime(netflix['date_added'].str.strip())
```

```
[27]: netflix.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 8807 entries, s1 to s8807
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   type             8807 non-null   object
1   title            8807 non-null   object
2   director         6173 non-null   object
3   cast             7982 non-null   object
4   country          7976 non-null   object
5   date_added       8797 non-null   datetime64[ns]
6   release_year     8807 non-null   int64
7   rating           8807 non-null   object
8   duration         8807 non-null   object
9   listed_in       8807 non-null   object
10  description      8807 non-null   object
dtypes: datetime64[ns](1), int64(1), object(9)
memory usage: 1.1+ MB
```

```
[28]: netflix.isna().sum()
```

```
[28]: type             0
      title           0
      director       2634
      cast           825
      country        831
      date_added     10
      release_year    0
      rating         0
      duration       0
      listed_in      0
      description    0
      dtype: int64
```

```
[29]: # Checking the rows which have missing date_added values
      netflix[netflix['date_added'].isnull()]
```

```
[29]:
```

	type	title	director	\
show_id				
s6067	TV Show	A Young Doctor's Notebook and Other Stories		NaN
s6175	TV Show	Anthony Bourdain: Parts Unknown		NaN
s6796	TV Show	Frasier		NaN
s6807	TV Show	Friends		NaN
s6902	TV Show	Gunslinger Girl		NaN
s7197	TV Show	Kikoriki		NaN
s7255	TV Show	La Familia P. Luche		NaN

s7407	TV Show	Maron	NaN
s7848	TV Show	Red vs. Blue	NaN
s8183	TV Show	The Adventures of Figaro Pho	NaN

	cast	country \
show_id		
s6067	Daniel Radcliffe, Jon Hamm, Adam Godley, Chris...	United Kingdom
s6175	Anthony Bourdain	United States
s6796	Kelsey Grammer, Jane Leeves, David Hyde Pierce...	United States
s6807	Jennifer Aniston, Courteney Cox, Lisa Kudrow, ...	United States
s6902	Yuuka Nanri, Kanako Mitsuhashi, Eri Sendai, Am...	Japan
s7197	Igor Dmitriev	NaN
s7255	Eugenio Derbez, Consuelo Duval, Luis Manuel Áv...	United States
s7407	Marc Maron, Judd Hirsch, Josh Brener, Nora Zeh...	United States
s7848	Burnie Burns, Jason Saldaña, Gustavo Sorola, G...	United States
s8183	Luke Jurevicius, Craig Behenna, Charlotte Haml...	Australia

	date_added	release_year	rating	duration \
show_id				
s6067	NaT	2013	TV-MA	2 Seasons
s6175	NaT	2018	TV-PG	5 Seasons
s6796	NaT	2003	TV-PG	11 Seasons
s6807	NaT	2003	TV-14	10 Seasons
s6902	NaT	2008	TV-14	2 Seasons
s7197	NaT	2010	TV-Y	2 Seasons
s7255	NaT	2012	TV-14	3 Seasons
s7407	NaT	2016	TV-MA	4 Seasons
s7848	NaT	2015	NR	13 Seasons
s8183	NaT	2015	TV-Y7	2 Seasons

	listed_in \
show_id	
s6067	British TV Shows, TV Comedies, TV Dramas
s6175	Docuseries
s6796	Classic & Cult TV, TV Comedies
s6807	Classic & Cult TV, TV Comedies
s6902	Anime Series, Crime TV Shows
s7197	Kids' TV
s7255	International TV Shows, Spanish-Language TV Sh...
s7407	TV Comedies
s7848	TV Action & Adventure, TV Comedies, TV Sci-Fi ...
s8183	Kids' TV, TV Comedies

	description
show_id	
s6067	Set during the Russian Revolution, this comic ...
s6175	This CNN original series has chef Anthony Bour...

```
s6796    Frasier Crane is a snooty but lovable Seattle ...
s6807    This hit sitcom follows the merry misadventure...
s6902    On the surface, the Social Welfare Agency appe...
s7197    A wacky rabbit and his gang of animal pals hav...
s7255    This irreverent sitcom featues Ludovico, Feder...
s7407    Marc Maron stars as Marc Maron, who interviews...
s7848    This parody of first-person shooter games, mil...
s8183    Imagine your worst fears, then multiply them: ...
```

All the rows having missing date_added column are from TV show categories. I am deciding to drop these rows as i think it will not affect the analysis too much.

```
[38]: netflix.dropna(subset=['date_added'], axis=0, inplace=True)
```

```
[39]: netflix.isna().sum()
```

```
[39]: type           0
      title         0
      director    2624
      cast        825
      country     830
      date_added   0
      release_year 0
      rating       0
      duration     0
      listed_in    0
      description  0
      dtype: int64
```

Hence we have cleared the data, removed missing values, corrected data types, and removed erroneous values in the data.

```
[ ]:
```

2.0.2 Analysis of top Countries where the netflix content is streaming

to find out the top countries, we need to see the countries column and count the number of times any country is appearing in the countries. The challenge is to count where there are multiple countries.

```
[43]: netflix.head(10)
```

```
[43]:
```

	type	title \
show_id		
s1	Movie	Dick Johnson Is Dead
s2	TV Show	Blood & Water
s3	TV Show	Ganglands
s4	TV Show	Jailbirds New Orleans
s5	TV Show	Kota Factory

s6	TV Show	Midnight Mass
s7	Movie	My Little Pony: A New Generation
s8	Movie	Sankofa
s9	TV Show	The Great British Baking Show
s10	Movie	The Starling

	director \
show_id	
s1	Kirsten Johnson
s2	NaN
s3	Julien Leclercq
s4	NaN
s5	NaN
s6	Mike Flanagan
s7	Robert Cullen, José Luis Ucha
s8	Haile Gerima
s9	Andy Devonshire
s10	Theodore Melfi

	cast \
show_id	
s1	NaN
s2	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...
s3	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...
s4	NaN
s5	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...
s6	Kate Siegel, Zach Gilford, Hamish Linklater, H...
s7	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...
s8	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...
s9	Mel Giedroyc, Sue Perkins, Mary Berry, Paul Ho...
s10	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...

	country	date_added \
show_id		
s1	United States	2021-09-25
s2	South Africa	2021-09-24
s3	NaN	2021-09-24
s4	NaN	2021-09-24
s5	India	2021-09-24
s6	NaN	2021-09-24
s7	NaN	2021-09-24
s8	United States, Ghana, Burkina Faso, United Kin...	2021-09-24
s9	United Kingdom	2021-09-24
s10	United States	2021-09-24

	release_year	rating	duration \
show_id			

s1	2020	PG-13	90 min
s2	2021	TV-MA	2 Seasons
s3	2021	TV-MA	1 Season
s4	2021	TV-MA	1 Season
s5	2021	TV-MA	2 Seasons
s6	2021	TV-MA	1 Season
s7	2021	PG	91 min
s8	1993	TV-MA	125 min
s9	2021	TV-14	9 Seasons
s10	2021	PG-13	104 min

show_id	listed_in \
s1	Documentaries
s2	International TV Shows, TV Dramas, TV Mysteries
s3	Crime TV Shows, International TV Shows, TV Act...
s4	Docuseries, Reality TV
s5	International TV Shows, Romantic TV Shows, TV ...
s6	TV Dramas, TV Horror, TV Mysteries
s7	Children & Family Movies
s8	Dramas, Independent Movies, International Movies
s9	British TV Shows, Reality TV
s10	Comedies, Dramas

show_id	description
s1	As her father nears the end of his life, filmm...
s2	After crossing paths at a party, a Cape Town t...
s3	To protect his family from a powerful drug lor...
s4	Feuds, flirtations and toilet talk go down amo...
s5	In a city of coaching centers known to train I...
s6	The arrival of a charismatic young priest brin...
s7	Equestria's divided. But a bright-eyed hero be...
s8	On a photo shoot in Ghana, an American model s...
s9	A talented batch of amateur bakers face off in...
s10	A woman adjusting to life after a loss contend...

```
[54]: def count_countries(df):
    countries = {}
    for country_list in df.country:
        print(type(country_list))
        if type(country_list) != float:
            for country in [country_name.strip() for country_name in
country_list.split(',')]:
                countries[country] = countries.get(country, 0) + 1

    return countries
```

```

[94]: # countries = count_countries(netflix)

[93]: # countries

[65]: top_countries = {country:countries[country] for country in countries.keys() if
    ↪countries[country] > 10}

[92]: # top_countries

[68]: top_countries = dict(sorted(top_countries.items(), key=lambda item: item[1],
    ↪reverse=True))

[70]: top_countries = pd.Series(data=top_countries.values(), index=top_countries.
    ↪keys())

[91]: # top_countries

[95]: def is_among_top_10(x, top_countries):
    if type(x) != float:
        for item in x.split(','):
            if item.strip() in top_countries[:11]:
                return True

    return False

top_10_movies_countries = netflix.loc[(netflix['type']=='Movie') &
    ↪(netflix['country'].apply(lambda x: is_among_top_10(x, top_countries)))]

top_10_movies_countries

```

```

[95]:
      type      title      director \
show_id
s1      Movie  Dick Johnson Is Dead  Kirsten Johnson
s8      Movie              Sankofa    Haile Gerima
s10     Movie    The Starling      Theodore Melfi
s13     Movie    Je Suis Karl  Christian Schwochow
s25     Movie              Jeans        S. Shankar
...      ...      ...      ...
s8800   Movie              Zenda    Avadhoot Gupte
s8803   Movie              Zodiac    David Fincher
s8805   Movie    Zombieland      Ruben Fleischer
s8806   Movie              Zoom      Peter Hewitt
s8807   Movie              Zubaan    Moez Singh

      cast \
show_id
s1      NaN

```

s8	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...
s10	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...
s13	Luna Wedler, Jannis Niewöhner, Milan Peschel, ...
s25	Prashanth, Aishwarya Rai Bachchan, Sri Lakshmi...
...	...
s8800	Santosh Juvekar, Siddharth Chandekar, Sachit P...
s8803	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...
s8805	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...
s8806	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...
s8807	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...

	country	date_added	\
show_id			
s1	United States	2021-09-25	
s8	United States, Ghana, Burkina Faso, United Kin...	2021-09-24	
s10	United States	2021-09-24	
s13	Germany, Czech Republic	2021-09-23	
s25	India	2021-09-21	
...	...	...	
s8800	India	2018-02-15	
s8803	United States	2019-11-20	
s8805	United States	2019-11-01	
s8806	United States	2020-01-11	
s8807	India	2019-03-02	

	release_year	rating	duration	\
show_id				
s1	2020	PG-13	90 min	
s8	1993	TV-MA	125 min	
s10	2021	PG-13	104 min	
s13	2021	TV-MA	127 min	
s25	1998	TV-14	166 min	
...	...	...	...	
s8800	2009	TV-14	120 min	
s8803	2007	R	158 min	
s8805	2009	R	88 min	
s8806	2006	PG	88 min	
s8807	2015	TV-14	111 min	

	listed_in	\
show_id		
s1	Documentaries	
s8	Dramas, Independent Movies, International Movies	
s10	Comedies, Dramas	
s13	Dramas, International Movies	
s25	Comedies, International Movies, Romantic Movies	
...	...	

s8800	Dramas, International Movies
s8803	Cult Movies, Dramas, Thrillers
s8805	Comedies, Horror Movies
s8806	Children & Family Movies, Comedies
s8807	Dramas, International Movies, Music & Musicals

show_id	description
s1	As her father nears the end of his life, filmm...
s8	On a photo shoot in Ghana, an American model s...
s10	A woman adjusting to life after a loss contend...
s13	After most of her family is murdered in a terr...
s25	When the father of the man she loves insists t...
...	...
s8800	A change in the leadership of a political part...
s8803	A political cartoonist, a crime reporter and a...
s8805	Looking to survive in a world taken over by zo...
s8806	Dragged from civilian life, a former superhero...
s8807	A scrappy but poor boy worms his way into a ty...

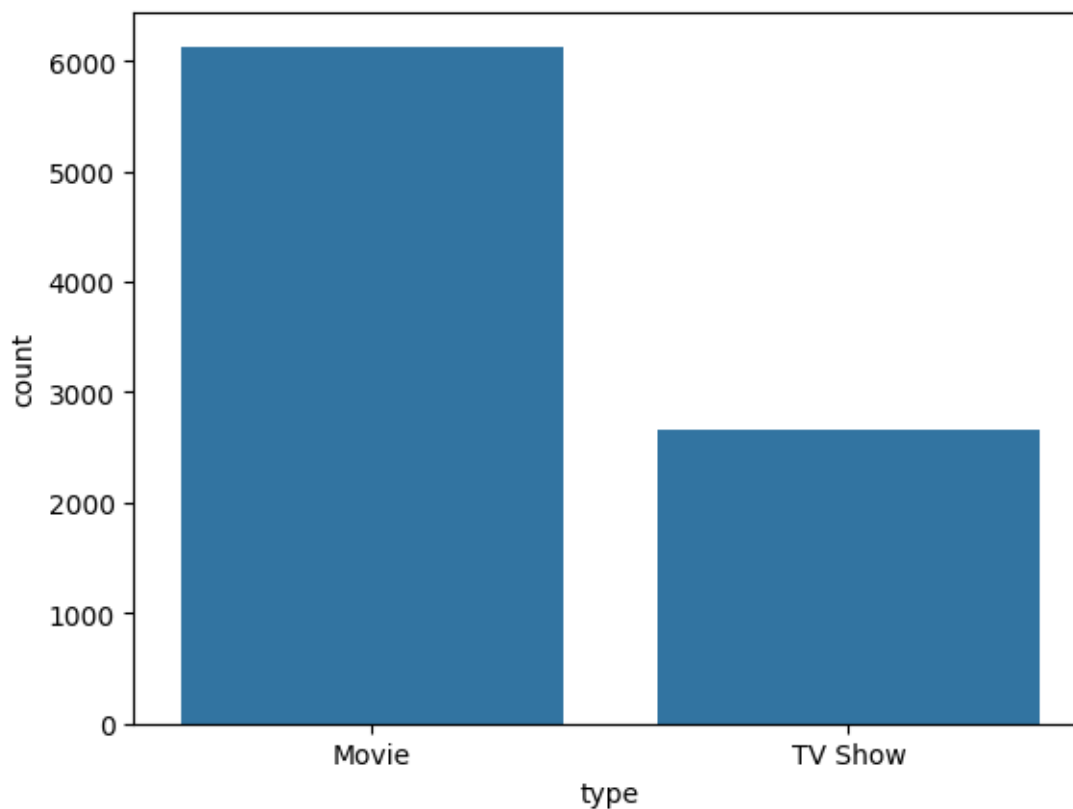
[4730 rows x 11 columns]

[]:

[]:

2.0.3 1.2 Comparison of Movies & TV shows on netflix

```
[99]: sns.countplot(data=netflix, x='type')
plt.show()
```



We can see that netflix has more focus on movies than TV Shows

2.0.4 1.3 How has the number of movies and TV Shows release changed over the years

```
[100]: # Bifurcation of Movies and TV shows for analysis
movies = netflix[netflix['type']=='Movie']
tv_shows = netflix[netflix['type']=='TV Show']
```

```
[103]: movies.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 6131 entries, s1 to s8807
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   type             6131 non-null   object
1   title            6131 non-null   object
2   director         5943 non-null   object
3   cast             5656 non-null   object
4   country          5691 non-null   object
5   date_added       6131 non-null   datetime64[ns]
```

```

6  release_year  6131 non-null  int64
7  rating        6131 non-null  object
8  duration      6131 non-null  object
9  listed_in     6131 non-null  object
10 description   6131 non-null  object
dtypes: datetime64[ns](1), int64(1), object(9)
memory usage: 574.8+ KB

```

```
[104]: tv_shows.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 2666 entries, s2 to s8804
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype
---  -
0   type            2666 non-null   object
1   title           2666 non-null   object
2   director        230 non-null    object
3   cast            2316 non-null   object
4   country         2276 non-null   object
5   date_added      2666 non-null   datetime64[ns]
6   release_year    2666 non-null   int64
7   rating          2666 non-null   object
8   duration        2666 non-null   object
9   listed_in       2666 non-null   object
10  description      2666 non-null   object
dtypes: datetime64[ns](1), int64(1), object(9)
memory usage: 249.9+ KB

```

```
[109]: movies_added_year = movies.groupby(movies['date_added'].dt.year).size()
```

```
[113]: tv_shows_released_year = tv_shows.groupby(tv_shows['date_added'].dt.year).size()
tv_shows_released_year
```

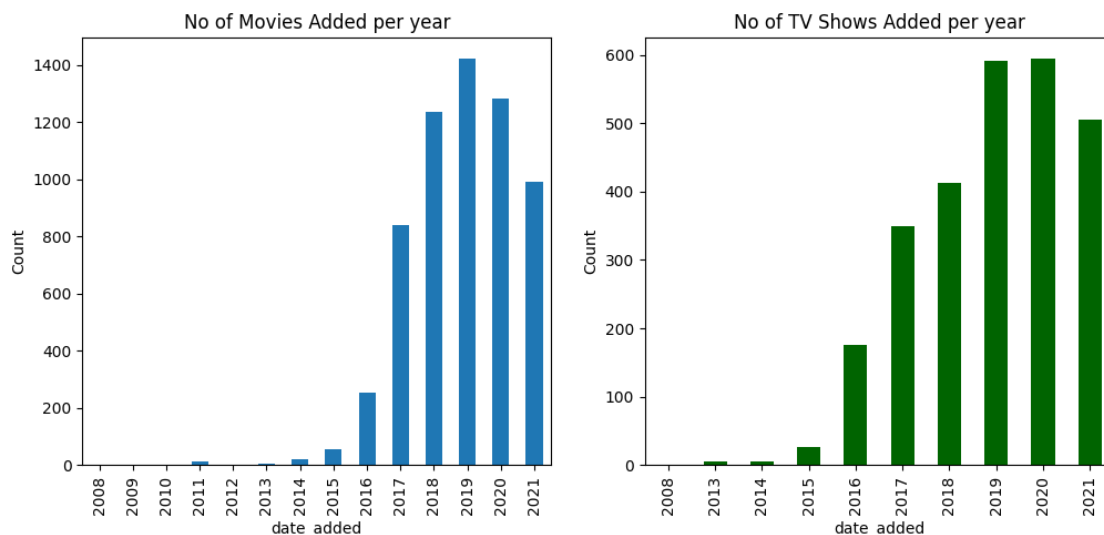
```

[113]: date_added
2008      1
2013      5
2014      5
2015     26
2016    176
2017    349
2018    412
2019    592
2020    595
2021    505
dtype: int64

```

```
[121]: plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
movies_added_year.plot(kind='bar')
plt.title('No of Movies Added per year')
plt.ylabel('Count')
plt.subplot(1, 2, 2)
tv_shows_released_year.plot(kind='bar', color='darkgreen')
plt.title('No of TV Shows Added per year')
plt.ylabel('Count')
plt.show()
```



Here we can see that the number of Movies and TV shows being added were increasing till 2019 and started decreasing after that

```
[ ]:
```

2.0.5 1.4 What is the best time to launch a TV show?

```
[128]: import calendar
month_order = list(calendar.month_name[1:])
month_order
```

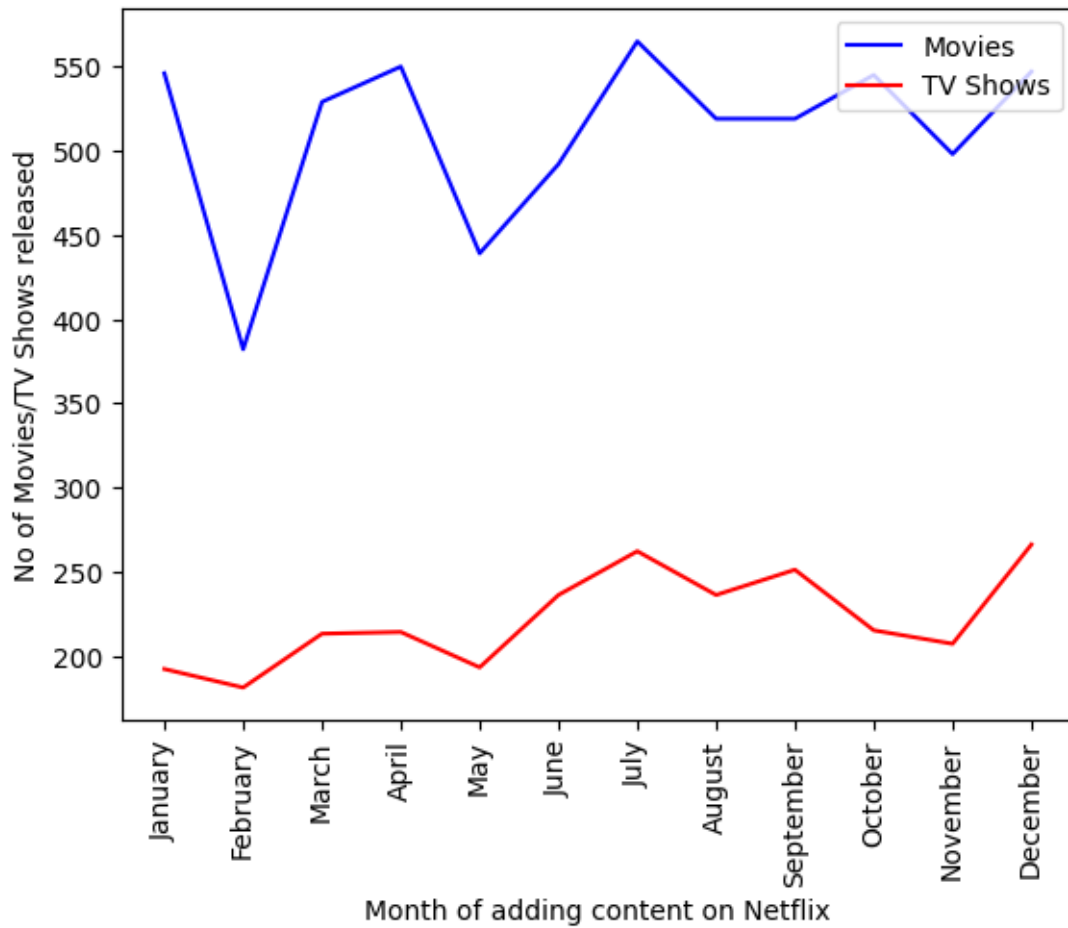
```
[128]: ['January',
        'February',
        'March',
        'April',
        'May',
        'June',
```

```
'July',  
'August',  
'September',  
'October',  
'November',  
'December']
```

```
[133]: tv_show_release = tv_shows.groupby(tv_shows['date_added'].dt.month_name()).  
      ↪size()  
      tv_show_release = tv_show_release.reindex(month_order, fill_value=0)  
      # tv_show_release
```

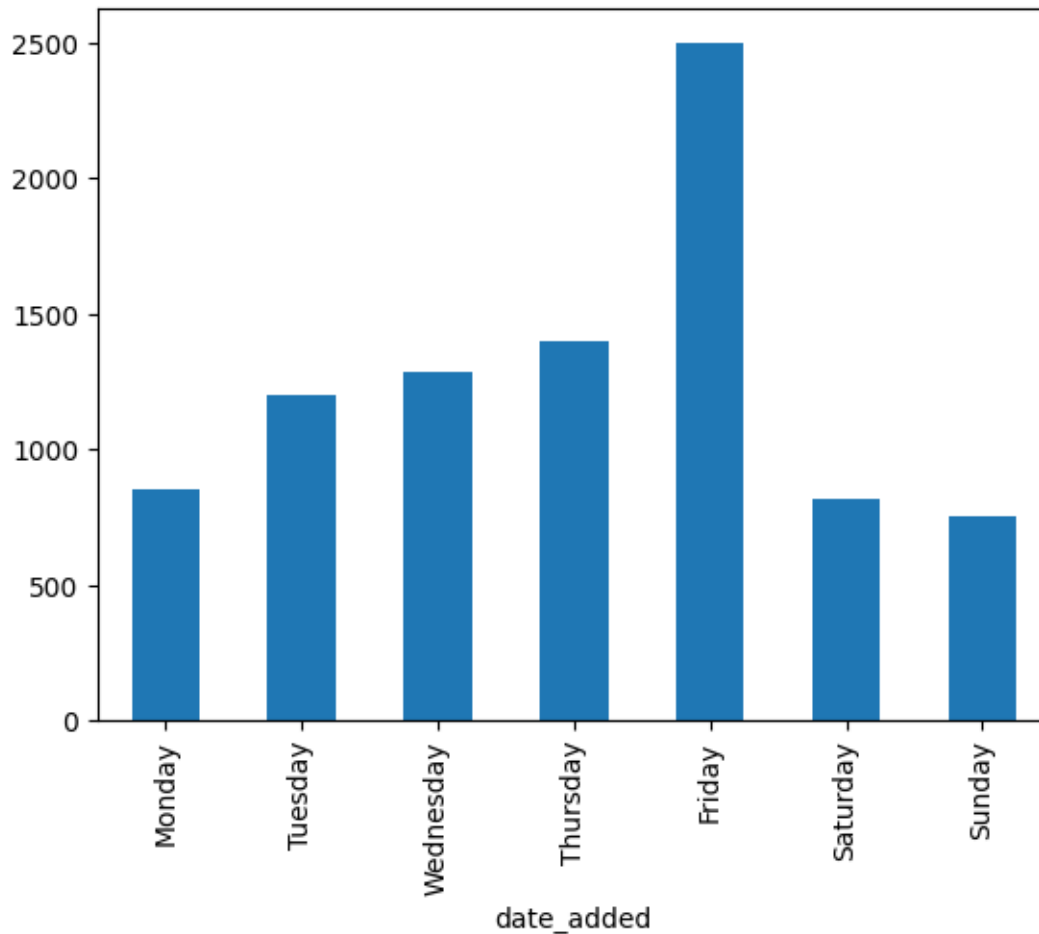
```
[134]: movie_release = movies.groupby(movies['date_added'].dt.month_name()).size()  
      movie_release = movie_release.reindex(month_order, fill_value=0)  
      # movie_release
```

```
[137]: plt.plot(movie_release.index, movie_release.values, color='b', label='Movies')  
      plt.plot(tv_show_release.index, tv_show_release.values, color='r', label='TV_  
      ↪Shows')  
      plt.legend(loc='upper right')  
      plt.xlabel('Month of adding content on Netflix')  
      plt.xticks(rotation=90)  
      plt.ylabel('No of Movies/TV Shows released')  
      plt.show()
```



```
[156]: # Identifying what is the best time of a week to release new content on netflix
netflix.groupby(netflix['date_added'].dt.day_name()).size().reindex(['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday'],
    fill_value=0).plot(kind='bar')
```

```
[156]: <Axes: xlabel='date_added'>
```



```
[141]: # The best time of the week to release new content is friday as it marks the
        ↪ beginning of the weekend.
```

We can clearly see a pattern here. So the best time to release any Movie/TV show during a year is July-August and December-January. This can be associated to the vacation and holiday season across the world. So it would benefit the viewership

2.0.6 1.5 Analysis of Top Genres in Movies and TV Shows

```
[142]: def count_genre(df):
        count = {}
        for genre in df.listed_in:
            for item in [x.strip() for x in genre.split(',')]:
                count[item] = count.get(item, 0)+1
        return count
```

```
[145]: movies_genre = count_genre(movies)
movies_genre_count = pd.Series(data=movies_genre.values(), index=movies_genre.
↳keys())
movies_genre_count
```

```
[145]: Documentaries      869
Children & Family Movies  641
Dramas                    2427
Independent Movies        756
International Movies      2752
Comedies                  1674
Thrillers                 577
Romantic Movies           616
Music & Musicals           375
Horror Movies             357
Sci-Fi & Fantasy          243
Action & Adventure        859
Classic Movies            116
Anime Features            71
Sports Movies             219
Cult Movies               71
Faith & Spirituality       65
LGBTQ Movies              102
Stand-Up Comedy           343
Movies                    57
dtype: int64
```

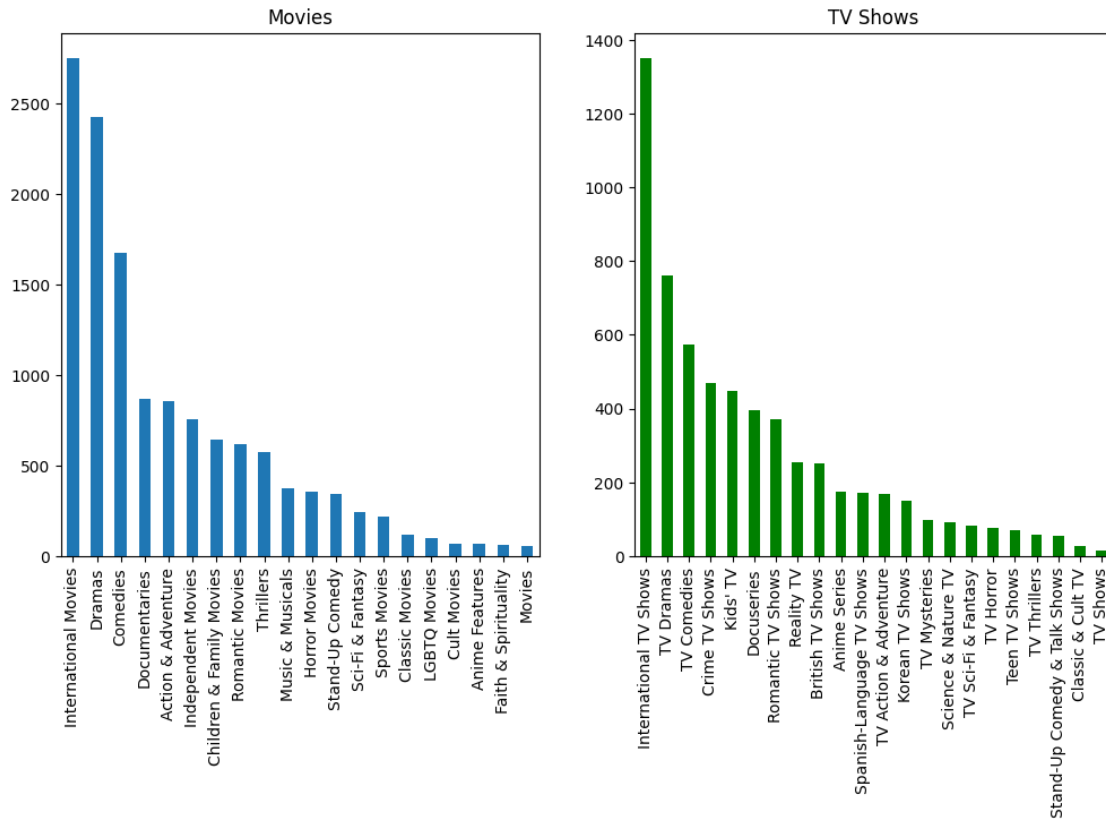
```
[146]: tv_genre = count_genre(tv_shows)
tv_genre_count = pd.Series(data=tv_genre.values(), index=tv_genre.keys())
tv_genre_count
```

```
[146]: International TV Shows  1350
TV Dramas                    762
TV Mysteries                 98
Crime TV Shows               469
TV Action & Adventure         167
Docuseries                   394
Reality TV                   255
Romantic TV Shows            370
TV Comedies                   574
TV Horror                     75
British TV Shows             252
Spanish-Language TV Shows    173
TV Thrillers                  57
Kids' TV                     449
TV Sci-Fi & Fantasy           83
Anime Series                  175
```

Korean TV Shows	151
Science & Nature TV	92
Teen TV Shows	69
TV Shows	16
Stand-Up Comedy & Talk Shows	56
Classic & Cult TV	26

dtype: int64

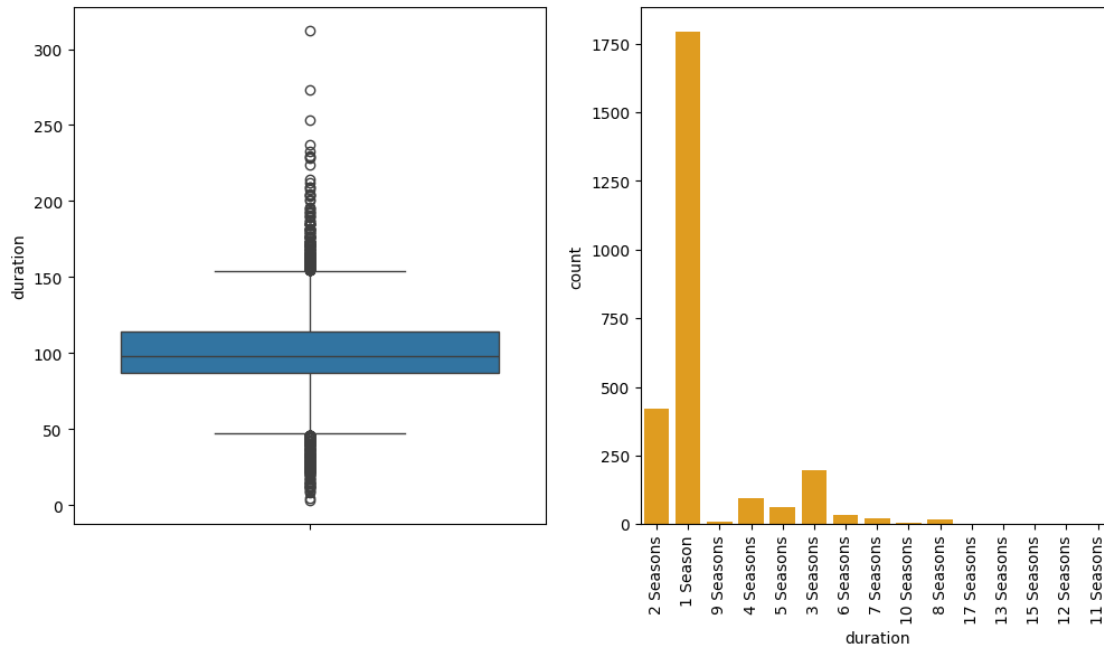
```
[155]: plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
movies_genre_count.sort_values(ascending=False).plot(kind='bar')
plt.title('Movies')
plt.subplot(1, 2, 2)
tv_genre_count.sort_values(ascending=False).plot(kind='bar', color='green')
plt.title('TV Shows')
plt.show()
```



Here We can see that the top are International Movies/TV Shows, Dramas, comedies, documentaries, crime, Action& Adventure etc. So the company can focus on these genre to increase veiwership

```
[164]: # # How does the duration of Movies and TV shows varies.
# movies['duration'] = movies['duration'].apply(lambda x: x.str[:4].
↳ astype(int))
```

```
[177]: fig, ax = plt.subplots(1, 2, figsize=(12, 6))
sns.boxplot(data=movies, y='duration', ax=ax[0])
sns.countplot(data=tv_shows, x='duration', ax=ax[1], color='orange')
plt.xticks(rotation=90)
plt.show()
```



```
[ ]:
```